

Minimax Dual Certificates for the Common-Action Obstruction

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Abstract

The common-action obstruction certifies that no single instrument serves every branch of an information set; its polyhedral form carries a Farkas multiplier (Abaee 2026a, Section 3.2). This paper states and proves the measure dual of the same obstruction: for control-affine dynamics with a compact convex control set, the common safe-action set is empty precisely when an adversarial probability measure on the active boundary–disturbance bundle drives the expected inward drift strictly negative against every control, with a certificate supported on at most $k+1$ points (k the control dimension; the bound is tight). The polyhedral case recovers the l_1 -normalized Farkas pair exactly; the singleton case recovers the Isaacs drift with the adversarial *distribution* over disturbances load-bearing. A two-action instance exhibits a strict minimax gap: without convexity the equivalence fails and no measure certifies — the audited finite codex systems sit outside this scope by design, certified combinatorially. An obstruction taxonomy (physical, epistemic, institutional) is instantiated on the audited grid, and a calibrated two-stock resource model — one fixed control set throughout — yields a closed-form critical aggregate $Y^* = \frac{27}{5}$: aggregate-yield fibres are viable exactly up to Y^* , the obstruction at $Y = 6$ is certified by a two-point dual measure with margin $\frac{3}{50}$, and the fibre-width formula $\Delta x_1(Y) = Y - 4$ gives the aggregation rule its exact reading. Dynamic-envelope, quantization, and stochastic-bridge extensions are recorded as conjectures; the residual dynamic-observation completeness gap of the source paper remains open. All claims are machine-verified in exact rational arithmetic (8 check families).

Introduction

The obstruction calculus’s certificates are emptiness statements: a belief is obstructed when a correspondence of jointly acceptable responses is empty (Abaee 2026a, Theorem 2; Abaee 2026b). For polyhedral safe-action sets the emptiness carries an explicit dual object — the Farkas multiplier vector — so that the obstruction itself becomes a checkable artifact; the sparse-witness bound (at most $m+1$ compatible states) and the singleton Isaacs drift complete the static picture. This paper unifies the static layer: the measure dual of the common-action obstruction,

of which the normalized Farkas pair, the sparse witness, and the Isaacs reduction are special cases. The measure dual is verified exactly; the sparsity bound is proved by a finite-reduction route (finite reduction, linear-programming duality, basic-solution support); the singleton Isaacs reading is stated with its load-bearing distributional caveat; the equality’s convexity boundary is exhibited by counterexample; the state-versus-epistemic separation is adopted as a working axiom; and a calibrated two-stock benchmark with a single fixed control set is solved in closed form. The dynamic layer — an adversarial-prior envelope for the restricted-information game — is then formulated and analyzed, and the residual dynamic-observation completeness gap (Abaee 2026a, Open Problem 1) is *not* closed here.

The measure dual of the common-action obstruction

Let the safe set be $V = \{x : q_j(x) \geq 0, j = 1, \dots, m\}$ with q_j of class C^1 ; let D be a compact-valued upper semi-continuous disturbance correspondence; let the control set $U \subseteq \mathbb{R}^k$ be compact and *convex*; and let the dynamics be affine in the control, $f(x, u, d) = f_0(x, d) + G(x)u$, with f_0 and G continuous. For a compact information set $B \subseteq V$ (a fibre or estimation tube) with nonempty active bundle

$$\mathcal{K}(B) = \{(x, j, d) : x \in B, q_j(x) = 0, d \in D(x)\},$$

define the inward drift $\psi(x, j, d; u) = \nabla q_j(x) \cdot [f_0(x, d) + G(x)u]$ and the expected drift $\Phi(u, \mu) = \int_{\mathcal{K}(B)} \psi d\mu$ over Borel probability measures μ on $\mathcal{K}(B)$.

Theorem 1 (minimax dual certificate). Under the standing hypotheses the common safe-action set $\mathcal{R}_V^B(B) = \bigcap_{x \in B} \mathcal{R}_V(x)$ is empty if and only if there exist an adversarial measure $\mu^* \in \mathcal{P}(\mathcal{K}(B))$ and a margin $\varepsilon > 0$ with

$$\max_{u \in U} \Phi(u, \mu^*) \leq -\varepsilon < 0. \quad (\star)$$

Moreover $\max_{u \in U} \min_{\mu \in \mathcal{K}(B)} \psi = \min_{\mu} \max_{u \in U} \Phi(u, \mu)$.

Proof. $\mathcal{P}(\mathcal{K}(B))$ is weak-* compact and convex (Banach – Alaoglu), and Φ is continuous, affine in u (control affinity), and linear in μ ; Sion’s minimax theorem applies, and the inner infimum over measures of a linear functional is attained by the Dirac measures, so $\max_u \min_{\mu} \Phi = \max_u \min_{\mu \in \mathcal{K}(B)} \psi = \min_{\mu} \max_u \Phi$. For the equivalence: $u \in \mathcal{R}_V^B(B)$ exactly when $\min_{\mu \in \mathcal{K}(B)} \psi(x, j, d; u) \geq 0$; the minimum is continuous in u , so

nonemptiness of the common safe-action set is equivalent to $\max_{u \in U} \min_{\mathcal{K}(B)} \psi \geq 0$ (the maximum is attained on the compact convex U), and emptiness is equivalent to the common value being strictly negative — which, by the equality, is equivalent to some μ^* with $\max_u \Phi(u, \mu^*) < 0$; the margin is the slack. $\square$

Proposition 1 (atomic support; $k + 1$ witnesses, tight). If $\mathcal{R}_V^B(B) = \emptyset$, the certificate μ^* may be chosen atomic with support of at most $k + 1$ triples, and the bound is tight in dimension k .

Proof. Write $\psi(x, j, d; u) = v(x, j, d) \cdot u + c(x, j, d)$ with $v = G(x)^\top \nabla q_j(x)$ and $c = \nabla q_j(x) \cdot f_0(x, d)$. On the compact set $U \times \mathcal{K}(B)$ the function $-\psi$ is continuous with $\min_u \max_{\mathcal{K}(B)} (-\psi) = 2\eta > 0$ (Theorem 1); a standard finite-cover argument extracts finitely many triples (x_i, j_i, d_i) , $i \leq N$, whose drift rows still satisfy $\min_{u \in U} \max_i (-\psi_i) \geq \eta$. Min-max duality for the finite system (linear programming duality over the simplex) yields $\lambda \geq 0$, $\sum_i \lambda_i = 1$, with $\sum_i \lambda_i \psi_i(u) < 0$ on U ; an extreme such λ is a basic solution of $k + 1$ active constraints and is supported on at most $k + 1$ indices. Tightness: in dimension $k = 2$ the rows $\{u_1 \geq 1\}$, $\{u_2 \geq 1\}$, $\{u_1 + u_2 \leq 1\}$ on the compact square $[-10, 10]^2$ are pairwise intersecting with empty total intersection (checked on the rational grid by the record), so no two witnesses suffice; for $k = 1$ the instance of E1 certifies with exactly two atoms. $\square$

The proposed argument for this proposition mixed Carathéodory’s theorem (which yields $k + 2$) with Helly’s; the route above — finite reduction, then duality, then the basic-solution support count — is the corrected one, and the record certifies tightness directly.

Proposition 2 (recoveries). (i) *Polyhedral*. If the drift rows are affine with rational data, the measure certificate is the ℓ_1 -normalized Farkas pair of Abaee 2026a, Section 3.2: on the two-floor instance $\{u \leq \frac{2}{5}\} \cap \{u \geq \frac{3}{5}\}$ with $U = [0, 1]$, the measure $\lambda = (\frac{1}{2}, \frac{1}{2})$ makes the expected drift *constant* $-\frac{1}{10}$, the normalized multiplier pair of the worked certificate in Section 3.4 there; on the feasible contrast $\{u \leq \frac{2}{5}\} \cap \{u \geq \frac{1}{5}\}$ both sides of Theorem 1 equal $+\frac{1}{10}$ and no certificate exists. (ii) *Singleton*. For $B = \{x\}$ the dual measures live on the disturbance set; on the exact convex instance $\psi(u, d_1) = 1 - 2u$, $\psi(u, d_2) = 2u - 1$, $\max_u \min_d \psi = \min_\mu \max_u \Phi = 0$, while the single-worst-disturbance reading $\min_d \max_u \psi = 1$ is *not* the dual: the adversarial distribution over disturbances is load-bearing, and the Isaacs drift is its Dirac-supported inner infimum.

Proposition 3 (convexity boundary). The equivalence of Theorem 1 fails without control convexity. With $U = \{0, 1\}$ and drift rows $(-1, +1)$, $(+1, -1)$, the common safe-action set is empty, yet $\max_u \min \psi = -1$ while $\min_\mu \max_u \Phi = 0$: a strict minimax gap, and no adversarial measure certifies. The audited finite action codices of the catalogue notes are non-convex by construction; there the obstructions are certified combinatorially, and the present theorem is a companion for the continuous, control-affine layer.

The obstruction taxonomy

The standing axiom adopted from the audit: *a dead state is never an information failure*. Three classes of nonviability are distinguished and, on the audited grid, are instantiated side by side (all exact): *physical* — the corner $(1, 1)$ under two floors is nonviable as a singleton under full information (no instrument keeps both floors); *epistemic* — the audited belief $\{(1, 2), (2, 1)\}$ has every branch viable and the joint set empty (failure by merge); *institutional* — under the unanimity protocol with agency 2 restricted to vote 1, the decentralized

kernel collapses from 12 to 0 (failure by protocol; full enumeration). The taxonomy disciplines attribution across the catalogue: mode-blind losses on the same grid are generated entirely at the singleton level, and no count that mixes the classes is admissible as an epistemic claim.

A calibrated two-stock benchmark

Leaving the audited grid, the continuous layer needs a worked system with one fixed control set throughout. Two renewable stocks ($r_1 = 1$, $r_2 = \frac{4}{5}$, $K_1 = K_2 = 10$, cross-competition $\frac{1}{20}$, shocks $|d_i| \leq \frac{1}{10}$), floors $x_i \geq 2$, and the single control set $U = \{u \geq 0 : u_1 + u_2 \geq 2\}$ (a demand floor; no capacity constraint is imposed). On the aggregate-yield fibre $Y = x_1 + x_2$ within the floors, the active-floor harvest caps are

$$\text{cap}_1(Y) = \frac{3}{2} - \frac{Y-2}{10}, \quad \text{cap}_2(Y) = \frac{59}{50} - \frac{Y-2}{10},$$

so the fibre is viable exactly when $\text{cap}_1(Y) + \text{cap}_2(Y) \geq 2$, i.e.

Theorem 2 (critical aggregate). $Y^* = \frac{27}{5}$: aggregate fibres with $Y \leq \frac{27}{5}$ are viable and those with $Y > \frac{27}{5}$ are obstructed, the value being exact ($\text{cap}_1 + \text{cap}_2 = 2$ at Y^*). At $Y = 5$ the fibre is viable with witness $u = (\frac{6}{5}, \frac{4}{5})$; at $Y = 6$ it is obstructed ($\text{cap sum } \frac{47}{25} < 2$) with dual certificate $\mu^* = (\frac{1}{2}, \frac{1}{2})$ on the two active-floor atoms — two points, within the $k + 1 = 3$ bound — and margin $\varepsilon = \frac{3}{50}$.

The reading is the opposite of the intuition that abundance certifies safety: cross-competition tightens the harvest caps as the aggregate grows, so richer fibres obstruct first — aggregate yield is a failing indicator in both directions. The fibre-width formula makes the aggregation rule exact: the Y -fibre within the floors is $x_1 \in [2, Y - 2]$, of width $\Delta x_1(Y) = Y - 4$, so a linear index cannot separate states closer together than the floor buffer — the composite index is certifying exactly when the width stays inside the buffer, and the certainly-safe reporting of the applications companion governs otherwise.

The dynamic envelope

A natural candidate formulation of a dynamic certificate suggests itself: an adversarial measure curve $t \mapsto \mu_t^*$ whose expected-margin envelope $Q(t)$ decreases at a certified rate, governed by an information-set operator acting on the state. The candidate fails, for three reasons that shape the formulation developed below; this section records the reasons, formulates the corrected object, and proves what exact methods reach. The residue remains conjectural.

Three obstructions to the candidate

(i) *A Bellman object posed as a pointwise Hamiltonian*. The candidate operator maps a value along trajectories to an expectation over an information set of measures — a Bellman-type, hence nonlocal, object — while dressed as a pointwise differential operator. Restricted-information games do not produce pointwise Hamiltonians in the state; their well-posed home is the information state (here, the belief), on which the object is a dynamic program. The nonlocality is not itself the obstruction; the pointwise differential form is.

(ii) *An inconsistent measure curve*. A curve $t \mapsto \mu_t^*$ of measures carries no requirement that μ_t^* be the marginal at time t of one measure on disturbance paths. Without that consistency the envelope has no time consistency: nothing connects $Q(t)$ to $Q(t')$ for $t' > t$, so no backward recursion and no comparison can even be posed. The formulation developed below replaces the curve by a single adversarial prior μ^* on disturbance paths;

conditional expectation then supplies the martingale structure without any PDE.

(iii) *Expectation versus trajectory.* For the certificate direction the selection argument is not needed at all: an expectation of a bounded functional that is strictly negative exhibits a positive-measure set of realizations on which the functional is strictly negative. Selection theorems are required only for the converse value-attainment direction, which the certificate does not use.

The corrected object

Lemma 1 (expectation-to-realization bridge). Let μ^* be a probability measure on disturbance paths compatible with the information structure, let π be nonanticipative, and let the safety functional $F^\pi(\omega)$ — the realized weighted facet margin at the horizon — be bounded below and $\mathcal{F}_T$ -measurable. If $\mathbb{E}^{\mu^*}[F^\pi] < 0$, then $\mu^*(\{\omega : F^\pi(\omega) < 0\}) > 0$: some compatible disturbance realization violates the margin, and the set of such realizations carries positive μ^* -measure.

Proof. Nonanticipativity of π makes F^π an $\mathcal{F}_T$ -measurable functional of the disturbance path, and boundedness below makes its expectation well-defined. If $\mu^*(F^\pi < 0) = 0$, then $F^\pi \geq 0$ holds μ^* -almost surely, so $\mathbb{E}^{\mu^*}[F^\pi] \geq 0$ — contradicting the hypothesis. The realizations of the positive-measure set are compatible by construction, since μ^* is supported on the compatible set. $\square$

Definition 1 (adversarial-prior envelope). Fix a horizon T , a bounded safety functional F (positive values safe), and one consistent adversarial prior μ^* on disturbance paths. The envelope is

$$Q(t) = \operatorname{ess\,sup}_{\pi} \mathbb{E}^{\mu^*}[F^\pi \mid \mathcal{F}_t],$$

the essential supremum over nonanticipative policies of the conditional expected safety functional. For each fixed π the process $t \mapsto \mathbb{E}^{\mu^*}[F^\pi \mid \mathcal{F}_t]$ is a martingale; the envelope is its essential supremum over policies, and its evaluation at information states — a Bellman recursion, not a differential operator — is the well-posed successor of the proposed information-set operator.

Theorem 3 (finite well-posedness, comparison, exactness). In the finite setting — disturbance set Ω finite, information states b drawn from a finite partition, admissible actions finite, data rational — define the one-step operator

$$\Phi V(b) = \max_a \mathbb{E}^{\mu^*}[V(b') \mid b, a],$$

where b' is the posterior given (b, a, y) the maximum over admissible actions of the prior-averaged next-belief value. Then: (i) Φ is order-preserving, $V \leq W \Rightarrow \Phi V \leq \Phi W$ pointwise; (ii) $Q(0) = \Phi^N g$ is computed exactly by backward induction, in rational arithmetic; and (iii) $Q(0) < 0$ certifies that for every admissible policy the set of disturbance realizations defeating it carries positive μ^* -measure.

Proof. (i) If $V \leq W$ pointwise, then for each (b, a) the conditional expectation of $V(b')$ is at most that of $W(b')$ (monotonicity of expectation over the same kernel), and the maximum over the same action set preserves the inequality. (ii) By induction on the remaining horizon: $\Phi^0 g = g$ is exact by definition; the induction step applies Φ once, and each evaluation is a finite weighted sum of previously computed values, so rational data propagate exactly through the recursion. Well-posedness — existence and uniqueness of the recursion's output — is supplied by the same induction. (iii) $Q(0) < 0$ bounds every policy's conditional expectation: for the optimal policy the expectation is $Q(0) < 0$, and for every suboptimal π the expectation is smaller still, so Lemma 1 applies to each π and yields the positive-measure defeating set. $\square$

Proposition 4 (identification with the recourse certificate). One step deep, with stored facet margins β_ℓ , control kernels $k_\ell(\sigma)$, prior λ over the stored branches, and control set U , the envelope evaluation of Definition 1 — the policy's optimal ceiling on the expected weighted facet sum — is exactly the support-function certificate

$$\Gamma_h(\lambda) = \sum_{\ell} \lambda_{\ell} \beta_{\ell} + \int_0^{\tau} h_U \left(\sum_{\ell} \lambda_{\ell} k_{\ell} \right) + \sum_{\ell} \int_{\tau}^T h_U(\lambda_{\ell} k_{\ell}),$$

with $h_U(g) = \max_{v \in U} g^\top v$ the support function of U , of the obstruction calculus (its Section 3.7; complete proof in its Supplementary Material, S1). The recourse certificate is the corrected dynamic envelope's one-step evaluation.

Proof. Variation of constants expands each realization of the weighted facet sum into the stored β -term plus the window pairing (one common signal on $[0, \tau)$) plus the branch pairings (separate post-reveal signals). The supremum of a linear functional over the signal sets factorizes across the time intervals, and each interval's supremum of a realized pairing $g^\top v$ over $v \in U$ is the support value $h_U(g)$; collecting terms gives exactly $\Gamma_h(\lambda)$ as the ceiling of every policy's realized weighted facet sum. Negativity of the ceiling certifies by Lemma 1: the realized weighted facet sum of every policy is at most $\Gamma_h(\lambda) < 0$, so some facet is violated on a positive-measure set of realizations. $\square$

Corollary 1 (the three-branch instance, certified). On the braking instance of the obstruction calculus (three branches, weights $\lambda = (\frac{3}{8}, \frac{5}{16}, \frac{5}{16})$, stored margin $\beta = \frac{16}{25} - \tau - 1$): the pooled window kernel vanishes ($\sum_{\ell} \lambda_{\ell} n_{\ell} = 0$), each branch's post-reveal pairing contributes its support value $\lambda_{\ell} \int_{\tau}^{\tau+1} (\tau + 1 - \sigma) d\sigma = \lambda_{\ell}/2$, and the envelope evaluation is $\Gamma_h(\tau) = \frac{7}{50} - \tau$: negative exactly for $\tau > \frac{7}{50} = 0.14$, where the bridge converts it into the positive-measure defeating set for every blind-window policy; hold-then-brake is safe exactly to $\tau = \frac{7}{50}$, so the certificate is tight. The verification script evaluates the identity exactly at $\tau \in \{\frac{1}{10}, \frac{7}{50}, \frac{1}{5}\}$.

Conjecture 1 (the open residue in continuous time). The continuous-time residue of the corrected formulation: (i) an evolution law for the adversarial prior along the review sequence — a consistent coupling of the one-step priors, in the spirit of martingale-transport couplings — under which the envelope obeys a dynamic-programming equation in continuous time; (ii) the limit of the information-state recursion under partition refinement, which is where a comparison principle must live if a differential statement is to be recovered at all; and (iii) value attainment (the selection direction) in the general filtered setting. The static identification of Proposition 4 is exact and is not part of the conjecture.

The review-sequence envelope: exact couplings

The envelope of the preceding section is one review deep. This section certifies the two properties its continuous-time residue must preserve — coupling consistency and refinement monotonicity — on exact review-sequence instances; everything is rational arithmetic, regenerated by the script.

The instance

Two windows, one hidden branch. The branch is $s \in \{+1, -1\}$, prior $\frac{1}{2}$ each; the state starts at $z_0 = 2$. At each review the policy acts $u_1, u_2 \in \{-2, 0, 2\}$; between reviews the adversarial disturbance acts $d_1, d_2 \in \{-1, 1\}$ (one consistent prior: the product measure, $\frac{1}{2}$ per value per window). The state obeys

$z_1 = z_0 + s u_1 + d_1$ and $z_2 = z_1 + s u_2 + d_2$; the review after window 1 observes z_1 exactly. The safety functional is the expected two-sided margin

$$F = \mathbb{E}_s[2 - |z_2(s)|], \quad z_2(s) = z_0 + s(u_1 + u_2) + d_1 + d_2,$$

bounded on the finite instance, so the bridge lemma applies verbatim.

Proposition 5 (consistent coupling and the tower identity). The one-step adversarial priors couple to the product measure on (d_1, d_2) , and for *every* nonanticipative policy on the instance — all $3^4 = 81$ policy trees over the reachable z_1 cells — the path-enumerated expectation and the iterated-conditional (tower) evaluation of $\mathbb{E}[F]$ agree exactly. Each fixed policy's conditional-expectation process along the review filtration is therefore a martingale, and the envelope $Q_0 = \text{ess sup}_\pi \mathbb{E}[F \mid \mathcal{F}_0]$ is computed by backward induction: $Q_0 = \Phi^2 g$ equals the policy-tree supremum ($= \frac{1}{2}$, at $u_1 = \pm 2$).

Proof. Consistency is by construction: the product measure has the one-step priors as its marginals. For the tower identity, condition on $\mathcal{F}_1$ (the review): within each z_1 cell the inner conditional expectation is a finite weighted average, and the outer average over the cells reconstructs the path sum because the cells partition the paths — an induction on the two-node filtration, exact in rationals. The martingale property is the tower identity restated along the filtration. Backward induction computes $\Phi^2 g$: Φg at each cell is a maximum of three rational expectations, and Φ is applied once more over u_1 ; the script checks the result against the enumeration of all 81 trees. $\square$

Proposition 6 (information design and recourse strictly pay). On the instance: the envelope value is $Q_0 = \frac{1}{2}$, attained by $u_1 = -2$ (any $u_1 = \pm 2$ by symmetry) with the recourse $u_2(z_1) = (0, -2, +2, +2)$ at $z_1 = (-1, 1, 3, 5)$; every open-loop policy and the do-nothing policy attain only 0. The strict gap $\frac{1}{2} > 0$ is the certified content of the correction: the first review's action *designs the information* ($u_1 = -2$ makes z_1 separate the branch exactly), and the second *exploits it* (brake toward the origin on the revealed branch); at $z_1 = 5$ the revealed state is unrecoverable and the cell value is -1 .

Proof. The four singleton cells at $u_1 = -2$ carry values $Q_1 = (1, 1, 1, -1)$ at weights $\frac{1}{4}$ each: $Q_0 = \frac{1}{2}$. The open-loop maximum is the better of nine fixed pairs, computed exactly as 0; the do-nothing pair $(0, 0)$ also evaluates to 0. The cell values and the maximizing u_2 are direct rational computations, reproduced by the script. $\square$

Proposition 7 (refinement is monotone). On the three-state matching instance — hidden $x \in \{1, 2, 3\}$, prior $\frac{1}{3}$ each, margin $-|x - a|$, action a per observation block — the envelope is non-decreasing under partition refinement and strictly increasing on the instance's final step: $Q(\{\{1, 2\}, \{3\}\}) = Q(\{\{1\}, \{2, 3\}\}) = -\frac{1}{3} < 0 = Q(\{\{1\}, \{2\}, \{3\}\})$.

Proof. Every policy measurable under a coarser partition is measurable under a refinement, so the supima are ordered. Strictness: under the fine partition the policy matches each state exactly ($a = x$, value 0); under either coarse partition two states share an action and the best shared action leaves one mismatch, value $-\frac{1}{3}$. This monotone, bounded family is the direction the partition-refinement limit of the residue conjecture must take. $\square$

The residue conjecture now reads sharper: the continuous-time coupling law must reduce, on review sequences with finite disturbance sets, to the product coupling certified here; and the information-state recursion must converge along refinements with the monotonicity certified here.

The general finite-sequence envelope

The instance of the previous section is the $K = 2$ case of a general statement. This section states the general theorem over finite review sequences and proves it in full; the two-window instance is the $K = 2$ case, and the general argument uses only finite combinatorics and the linearity of expectation.

Setup

Fix a number of windows $K \geq 1$. Each window k carries a finite disturbance set D_k ; the disturbance paths form $\Omega = D_1 \times \dots \times D_K$, a finite set. A *consistent adversarial prior* is a product measure $\mu^* = \mu_1^* \otimes \dots \otimes \mu_K^*$ on Ω ; consistency — each marginal μ_k^* is the time- k law of the one measure μ^* — is exactly the requirement whose absence makes a backward recursion ill-posed. Reviews divide the sequence: the review filtration is an increasing sequence of partitions $\mathcal{P}_1 \preceq \mathcal{P}_2 \preceq \dots \preceq \mathcal{P}_K$ of Ω , each $\mathcal{P}_k$ consisting of the cells of the information available at review k (so $\mathcal{P}_k$ refines into $\mathcal{P}_{k+1}$), with $\mathcal{P}_K$ the singleton partition; $\mathcal{P}_1$ may be the trivial partition (a blind first window). Window k has a finite action set A_k ; a *nonanticipative policy* is $\pi = (\pi_1, \dots, \pi_K)$ with π_k constant on the cells of $\mathcal{P}_k$ — the filtration carries the whole observation history, so cell-constancy is adaptedness. The data are completed by a bounded safety functional $G : \Omega \times A_1 \times \dots \times A_K \rightarrow \mathbb{R}$ (positive values safe, per the sign convention of this paper), rational throughout; the realized functional of π is $G^\pi(\omega) = G(\omega, \pi_1[\omega]_1, \dots, \pi_K[\omega]_K)$, where $[\omega]_k$ is the $\mathcal{P}_k$ -cell of ω .

Theorem 4 (general finite-sequence envelope). Under the setup above: (i) *tower and martingale*. For every nonanticipative policy π ,

$$\mathbb{E}^{\mu^*}[G^\pi] = \mathbb{E}^{\mu^*}[\mathbb{E}^{\mu^*}[\dots \mathbb{E}^{\mu^*}[G^\pi \mid \mathcal{P}_K] \dots \mid \mathcal{P}_2] \mid \mathcal{P}_1],$$

and $M_k^\pi = \mathbb{E}^{\mu^*}[G^\pi \mid \mathcal{P}_k]$ is a μ^* -martingale along the filtration. (ii) *Dynamic programming*. Define the node values N on the review tree — nodes are pairs (cell, action prefix) — by $N_K(\{\omega\}, a_{1:K-1}) = \max_{a \in A_K} G(\omega, a_{1:K-1}, a)$ and, for $k < K$, the sum running over the children $C' \subseteq C$ of $\mathcal{P}_{k+1}$ with the child weights $w(C') = \mu^*(C')/\mu^*(C)$,

$$N_k(C, a_{1:k-1}) = \max_{a \in A_k} \sum_{C'} w(C') N_{k+1}(C', a_{1:k-1}a),$$

Then the envelope value $Q_0 := \sup_\pi \mathbb{E}^{\mu^*}[G^\pi]$ equals N_1 evaluated on the $\mathcal{P}_1$ -cells, averaged by their prior weights $Q_0 = \sum_{C \in \mathcal{P}_1} \mu^*(C) N_1(C)$ (a single node if $\mathcal{P}_1$ is trivial); the supremum is attained by a nonanticipative policy read off the maximizing arguments of N . (iii) *Refinement monotonicity*. If a second filtration $\mathcal{P}'$ refines $\mathcal{P}$ stage-wise, then the induced envelope values satisfy $Q_0(\mathcal{P}) \leq Q_0(\mathcal{P}')$. (iv) *Exactness and the obstruction direction*. All node values are rational; if $Q_0 < 0$, then for *every* nonanticipative π the set $\{\omega : G^\pi(\omega) < 0\}$ carries positive μ^* -measure.

Proof. (i) Each G^π is $\mathcal{P}_K$ -measurable (a finite function of the path and the cell-constant actions), and repeated conditioning is an identity on finite probability spaces; the tower formula is that identity read backwards, and the martingale property is $M_k^\pi = \mathbb{E}[M_{k+1}^\pi \mid \mathcal{P}_k]$, which is the defining property of conditional expectation applied twice. (ii) By backward induction on the stage. For $k = K$ the node value is the best terminal action, and G^π at a singleton cell is $G(\omega, \cdot)$ evaluated at the realized prefix and π_K , so $\max_{a \in A_K} G = N_K$ pointwise. For the step, fix a $\mathcal{P}_k$ -cell C and an action prefix, and split the policies by their stage- k action: every policy compatible with the prefix is a choice $a \in A_k$ together with a continuation policy for the

conditioned sub-instance on $\{C\}$ -successor cells, whose filtration $(\mathcal{P}_j)_{j>k}$ restricted to the successors refines to singletons at K . By (i) applied at the conditioning,

$$\mathbb{E}^{\mu^*}[G^\pi \mid C] = \sum_{C' \subseteq C} \frac{\mu^*(C')}{\mu^*(C)} \mathbb{E}^{\mu^*}[G^\pi \mid C'],$$

the conditional law of the successor cell given C is the ratio of prior masses (exogeneity: the prefix does not move μ^*), and the induction hypothesis — applied to each conditioned sub-instance, which has one fewer window — identifies the supremum of the inner conditional expectations over continuations of $(a_{1:k-1}, a)$ with $N_{k+1}(C', a_{1:k-1}a)$. Taking the outer maximum over a gives N_k ; summing over the $\mathcal{P}_1$ -cells with their prior weights and applying (i) once more gives $Q_0 = \sum_C \mu^*(C)N_1(C)$. Attainment: each N node is a maximum over a finite set; selecting a maximizing argument at every node and pasting along the tree yields a nonanticipative policy — cell-constancy holds because the selection is made per node — and its expectation is exactly the recursion’s value. (iii) If $\mathcal{P}'$ refines $\mathcal{P}$ stage-wise, every $\mathcal{P}$ -cell-constant policy is $\mathcal{P}'$ -cell-constant, so the $\mathcal{P}'$ -supremum ranges over a superset of policies and cannot be smaller; the inequality for the full sequences follows stage-wise from the one-step inclusion. (iv) The recursion’s only operations are maxima and μ^* -weighted averages of rationals, so every N is rational. If $Q_0 < 0$, then by (ii) every policy satisfies $\mathbb{E}^{\mu^*}[G^\pi] \leq Q_0 < 0$, and the expectation-to-realization bridge (the lemma of the corrected formulation) converts strict negative expectation into a positive-measure set of strictly negative realizations. $\square$

Remark 1 (the recourse certificate and the residue). Three consequences. *First*, the identification with the recourse certificate is the one-step case of Theorem 4: with stored facet margins the node recursion collapses to the support-function evaluation Γ_h (the identification proposition of the corrected formulation), so the calculus’s recourse certificate is the envelope’s K -node tree read at depth one. *Second*, the one-step priors μ_k^* do not determine the coupling μ^* : two consistent couplings with identical one-step marginals can give different node values (the script certifies a two-window example where the product coupling yields 0 and the comonotone coupling yields +1 on the same functional). The residue conjecture’s “evolution law for the adversarial prior” is therefore precisely a *coupling-selection* problem — choose the consistent coupling as a function of the state and review history — which is the finite-space form of the martingale-transport structure recorded in the conjecture. *Third*, refinement monotonicity (iii) is the finite-space grounding of the partition-limit clause: the instance family of the previous section, monotone and bounded, now carries a general ordering, and the continuous-time residue is exactly the question of which couplings and which refinement limits survive the passage to continuous disturbance sets.

What else is not claimed

Declarations

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